Task 8 — Docker Logs

DEVOPS-006: Docker Fundamentals & Containerization | Time: 45 Minutes

Step 1: Run the container

```
sudo docker run -d --name web-server nginx
```

This pulls (if not already present) and starts an Nginx container in detached mode (-d), naming it web-server.

Step 2: Check the logs

```
sudo docker logs web-server
```

This shows the Nginx startup logs — mostly config messages, since Nginx doesn't get real traffic logs until someone hits it.

Step 3: Follow the logs live

```
sudo docker logs -f web-server
```

The -f flag "follows" the log output in real time, similar to tail -f. Any new requests or events show up instantly as they happen.

Step 4: Stop following

Press CTRL + C to exit the live log stream (this doesn't stop the container, just detaches from the log stream).

Step 5: Inspect the container

```
sudo docker inspect web-server
```

This dumps a big JSON block with all the container's low-level details.

Step 6: Write down the key details

Field	Where to find it / meaning
Container ID	e.g. a1b2c3d4e5f6 — found near the top as "Id"
Image	nginx — the image it was built from, under "Image" or "Config" → "Image"
Status	e.g. running — under "State" → "Status"
Network information	IP address, gateway, MAC address — under "NetworkSettings"
Created time	the timestamp the container was created — under "Created"

What I learned: docker logs is useful for checking what's happening inside a running container without having to shell into it, and -f is handy for watching logs live. docker inspect gives the full metadata picture — good for debugging networking issues or confirming exactly what image/config a container is running.